Usage and resource accounting: independent Hutko reanalysis

Measured local execution and explicit remote-usage scenarios

13 October 2026

1 What is observed

The orchestration goal counter reported 686,232 tokens at the saved snapshot, after 1.31 hours of goal elapsed time. The exact snapshot is in `logs/agent-usage-snapshot.json`. This is an orchestration counter, not a billing export. It does not separate input, cached input, output, hidden reasoning, or tool billing; the short setup before goal creation and subsequent reporting tokens are not included. No exact provider charge can be recovered from it. The code itself makes no model API calls. Its normal execution therefore consumes zero inference tokens. Remote tokens belong to research, implementation, debugging, and writing by the assistant, not to subsequent scientific reruns. The counter is more defensible than estimating tokens from source-code characters, but its billing semantics are not exposed here. No claim of an exact dollar bill is made.

2 Dollar estimates with explicit assumptions

Official pricing distinguishes input, cached input, and output; it also depends on model, context length, and service tier (https://developers.openai.com/api/docs/pricing, consulted 13 October 2026). The exact billed model/tier, cache fraction, and account plan for this session are not available. The following rate scenarios are illustrative API equivalents, not assertions about the user's charge. Let N be the saved counter. Treating 80–95% of N as input and 5–20% as output gives

$$C = \frac{N_i[(1 - f_c)r_i + f_c r_c] + N_o r_o}{10^6}.$$

Here f_c is the cached fraction of input. We show zero caching as a conservative simple scenario; subscriptions, cache discounts, tool fees, and context premiums can change the amount substantially.

Illustrative rates per million tokens	Input USD	Output USD	Estimated USD range
Low	1.00	5.00	0.82–1.24
Middle	2.00	10.00	1.65–2.47
High	10.00	50.00	8.23–12.35

The token split is an assumption, not a measurement. The rate scenarios span published API price scales but do not identify the model that ran this task. An account usage/billing export is required for an actual charge.

3 Remote computing: scenarios rather than telemetry

No remote GPU count, active inference time, processor power, or memory measurement is exposed to this task. Elapsed assistant work includes local computations and network waits, and cannot be equated to

GPU active time. A rough planning envelope is therefore parameterized by 1–8 accelerators, 10–100% active duty, 300–1000 W per active accelerator, and 20–160 GB memory per accelerator. These are explicitly assumed engineering scenarios, not provider-specific measurements or inferred model architecture. Host CPU scenarios use 1–16 active cores and 32–512 GB host memory. Memory reservation is not the same as memory actually touched. Applied to 1.31 hours of elapsed work, these assumptions span 0.13–10.50 GPU-hours, 0.13–21.01 CPU-hours, 20–1280 GB accelerator memory, and 32–512 GB host memory. Accelerator energy alone spans 0.039–10.504 kWh. Cooling, networking, shared-server allocation, and model training are excluded. These wide ranges should not be cited as measured energy for this project.

4 Actual local execution

The build runner samples process-tree resident memory every 0.5 s and measures child-process user plus system CPU time. CPU-hours sum across cores; wall time does not. Summed RSS can count shared pages more than once. Peak memory is a sampled upper accounting measure, not an exact unique-working-set measurement. The machine exposes 96 logical CPUs and approximately 810 GB total RAM; this capacity is not the amount used by the calculation. The processing uses up to 16 worker processes, each with one BLAS thread. Migration is vectorized, with three workers per independent dataset and up to nine event workers. Local GPU use is zero GPU-hours. Across the instrumented build stages retained in the logs, local use totals 1.043 CPU-hours and 0.421 summed stage wall-hours. The maximum sampled process-tree RSS is 3.692 GB. The data directory contains 1.457 GB, including raw archive records. Multiple recorded builds are counted as actual repeated work, not combined into an estimate for a single rerun. Interactive development commands, the earlier installation/download commands, and final usage-document compilation are not fully included in these totals. GNU time logs for early commands are preserved separately.

Stage	Wall s	CPU s	Peak tree GB	Exit
process	318.3	1129.6	3.692	0
validate	6.0	5.9	0.258	0
process	24.4	230.7	3.117	0
validate	6.2	5.7	0.265	0
migrate	69.7	213.3	2.156	0
experiments	44.1	43.5	0.312	0
sensitivity	16.7	16.2	0.299	0
point-tests	16.8	16.4	0.275	0
compare	2.8	2.6	0.137	0
panels	0.6	0.1	0.012	0
plot	264.7	262.4	0.734	0
report-source	1.1	0.8	0.056	0
compile	5.6	5.2	0.037	0
process	24.3	228.8	3.107	0
validate	6.2	5.6	0.266	0
migrate	177.6	510.7	2.781	0
experiments	42.5	42.0	0.328	0
sensitivity	16.2	15.8	0.290	0
point-tests	16.8	16.4	0.278	0
compare	3.4	2.9	0.138	0
panels	0.6	0.1	0.012	0
plot	127.5	259.8	1.883	0
report-source	1.1	0.7	0.055	0

Stage	Wall s	CPU s	Peak tree GB	Exit
compile	6.1	5.6	0.037	0
process	24.2	228.5	3.109	0
validate	6.1	5.6	0.258	0
migrate	17.3	29.9	1.843	0
experiments	43.5	43.1	0.321	0
sensitivity	44.5	88.0	0.979	0
point-tests	22.3	48.3	0.967	0
compare	3.3	2.9	0.138	0
panels	0.6	0.1	0.012	0
plot	146.8	281.1	1.858	0
report-source	1.1	0.7	0.055	0
compile	5.6	5.2	0.038	0

Separate early GNU time logs (acquisition, an earlier preprocessing run, and the historical pilot migration) add 1.003 CPU-hours and 0.202 wall-hours. These commands predate the stage runner and do not overlap its recorded commands. Together the two measurement sources account for at least 2.046 CPU-hours; unmetered development commands make this a lower bound on all local work. GNU time's maximum RSS is not a sum across simultaneous children, so it is not substituted for the process-tree peak above. No local power meter or reliable package-energy counter was available. For a rough 100–500 W allocated-node envelope, multiply the actual stage wall-hours by 0.1–0.5 kW. This is a power-allocation assumption, especially on a shared 96-core machine, not a measured marginal draw. CPU time alone does not determine energy. At an assumed electricity rate of USD 0.10–0.30/kWh: the instrumented stages imply 0.0421–0.2103 kWh and USD 0.0042–0.0631 in electricity. Hardware amortization and archive service costs are excluded.

5 How to measure another run

Run `bash rebuild.sh`. Each stage produces a JSON record with wall seconds, CPU seconds, peak process-tree RSS, and return status. For a fresh numerical run, remove generated processed/image/validation outputs while retaining the raw archive cache, as documented in the README. Hardware power should be measured separately with a node power meter or available energy counters. Update the agent token snapshot only from genuine orchestration or billing telemetry; do not replace unknown values with asserted measurements.